BUSINESS INTELLIGENCE LAB

EXPERIMENT 6

Dashboard and Scorecard Development using Microsoft Power BI

Problem file

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1. Aim

To design and develop an interactive Executive Dashboard and Business Scorecard using Microsoft Power BI for monitoring organizational performance, supporting business decision-making, and communicating key performance indicators (KPIs) effectively.

2. Problem Statement

Global Superstore is a retail business that generates a large amount of sales and customer data. The executive management team requires a consolidated dashboard to monitor the organization's overall performance and make effective business decisions. The management wants to analyze important business metrics such as total sales, total profit, number of orders, profit margin, regional sales performance, category-wise sales, monthly sales trends, customer contribution, and segment-wise profitability.

The task is to develop an interactive Executive Dashboard and Business Scorecard using Microsoft Power BI. The dashboard should display important KPIs through suitable visualizations and allow users to filter the data based on Region, Category, and Year.

Additionally, a Business Scorecard must be created to compare actual performance against predefined targets for Sales, Profit, and Orders. Based on the dashboard findings, important business insights should be identified and recommendations should be provided to improve organizational performance.

3. Business Requirements

The following requirements must be fulfilled:

1. Develop an interactive Executive Dashboard using Microsoft Power BI.
2. Create KPI Cards for:
    - Total Sales
    - Total Profit
    - Total Orders
    - Profit Margin (%)
3. Create the following visualizations:
    - Sales by Region
    - Sales by Category
    - Monthly Sales Trend
    - Top 10 Customers by Sales
    - Profit by Segment
4. Add interactive slicers for:
    - Region
    - Category
    - Year
5. Create a Business Scorecard containing:
    - Sales Target vs Actual Sales
    - Profit Target vs Actual Profit
    - Order Target vs Actual Orders
6. Apply conditional formatting to represent performance status.
7. Identify the best-performing and worst-performing regions.
8. Identify the most profitable and least profitable categories.
9. Provide at least one business recommendation based on analytical findings.

4. Tools / Software Required

1. Microsoft Power BI Desktop
2. Global Superstore Dataset.xlsx
3. Power Query Editor
4. DAX (Data Analysis Expressions)

5. Dataset Description

Dataset Name: Global Superstore Dataset

Dataset Source: Kaggle

Dataset Link: https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset

The Global Superstore dataset contains business-related information about orders, sales, profits, customers, regions, product categories, and segments. It is used to analyze business performance and generate meaningful insights through Power BI visualizations.

6. Expected Output

The expected output of this experiment is an interactive Executive Dashboard and Business Scorecard containing:

- KPI cards displaying important business metrics.
- Charts representing regional, category-wise, monthly, customer-wise, and segment-wise performance.
- Interactive slicers for dynamic data filtering.
- A Business Scorecard comparing actual values with predefined targets.
- Performance indicators showing whether targets are achieved, partially achieved, or not achieved.
- Business insights and recommendations based on dashboard analysis.

9. Result

The problem requirements for developing an interactive Executive Dashboard and Business Scorecard using Microsoft Power BI were identified and documented. The experiment focuses on visualizing key business performance indicators, comparing actual performance with predefined targets, and generating meaningful business insights to support effective decision-making.